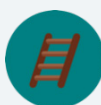


10.H1 Constructing Tangent Lines

Lesson Introduction

 Benchmark	MA.912.GR.5.5 Given a point outside a circle, construct a line tangent to the circle that passes through the given point.		 Learning Target	I can construct a line tangent to a circle that passes through a given point outside the circle.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	How do you construct a line tangent to a circle that passes through a given point outside the circle?			
 Lesson Narrative	Students will learn how to construct a line tangent to a circle that passes through a given point outside the circle with a compass and a ruler. Students will begin by collaborating to learn about tangent lines of circles and the steps to constructing tangent lines.			
 Component Summary	10.H.1 Warm-Up (~5 min.)	Determining radio signal distance and the signal's relationship to the Earth (<i>SE p. 170</i>)	10.H.3 Your Turn! (10-15 min.)	Constructing a tangent line and examining examples of constructions (<i>SE p. 172</i>)
	10.H.2 Collaboration (20-25 min.)	Constructing a tangent to a circle (<i>SE p. 170</i>)	10.H.4 Wrap-Up (~5 min.)	Constructing a tangent to a circle (<i>SE p. 173</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Glossary:</u> Tangent Line		- Compass - Ruler (Optional) Digital Version of Construction	
			Additional Preparation	
			This lesson is designated for students enrolled in the honors version of Geometry only.	

 Teacher Tips	Common Misconceptions - Students may skip a step in the construction process. - Students may think any line that intersects a circle in one or more points is a tangent.	Things to Consider Consider a refresher on the use of the tools and steps as part of the launch of the lesson.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 10.H.2 Collaboration provides students a collaborative opportunity to work together with constructing tangent lines. Students will have the opportunity for individual thinking and practice before collaborating with a partner to share strategies and results.	MA.K12.MTR.1.1: Participate in Effortful Learning 10.H.2 Collaboration provides students an opportunity to stay engaged through an interactive lesson with constructions.
 Differentiate Instruction	Consider showing students a digital version of the step-by-step construction process. This will allow entry points for visual and kinesthetic learners. Engage the class in a discussion about the steps taken in the construction to allow for an auditory entry point.	
Lesson Notes:		

10.H1.1 Warm-Up: Radio Signals

Component Overview: Students will explore the distance a radio signal would go based on their height.



10.H1.1 Warm-Up: Radio Signals

- Two-way radio communications help climbers stay in touch with base camp and contact the other expeditioners climbing Mt. Everest. The separate expeditions reserve a central channel where the various groups can share information on weather conditions and route changes or call for rescue.

The equation shows the maximum theoretical range of a radio signal, in miles, using a line-of-sight calculation, which considers the curvature of the Earth and presumes the Earth to be a perfect sphere – which of course it is not. This also assumes the radio signal is transmitted with sufficient power that maintains enough energy to travel the distance.

$$\text{Horizon(mi)} = 1.23 \times \sqrt{\text{height(ft)}}$$



- Use the formula to find the distance the radio signal would go based on your height.

Answers will vary.

$$5\text{ft } 6\text{in} = 5.5\text{ft} \quad 1.23 \times \sqrt{5.5} = 2.885 \text{ miles}$$

- Explain the relationship of the radio signal range (red line) and Earth.

Answers may vary.

The radio signal range is similar to a tangent line that touches the Earth at only one point.



Why would a lower height affect the path of the radio signal?

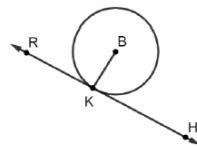
10.H1.2 Collaboration: Tangent Lines of Circles

Component Overview: Students will work together to construct tangent lines of circles. For this activity, students will need a compass and a ruler.



10.H1.2 Collaboration: Tangent Lines of Circles

1. Circle B is shown, with point K on the circle and $\overleftrightarrow{RK}$ tangent to circle B .



- a. Write two facts about circle B and tangent line $\overleftrightarrow{RK}$.
Answers will vary.

1. *Line RK intersects circle B at one point*
2. *Line RK is perpendicular to the radius BK*

- b. Share your facts with your partner and write down one additional fact.

3. *Answers will vary*
 $\angle BKH$ and $\angle BKR$ are right angles.

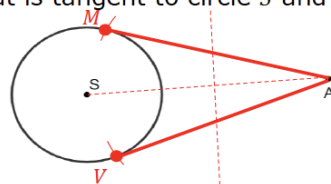
2. Complete the statement.

A line tangent to a circle touches a circle at

point(s), and the line is ☐ parallel ☒ perpendicular to the radius of the circle drawn to the point of tangency.

- ☐ one
- ☐ two
- ☐ three

3. To construct a line tangent to a circle, you need a compass and a ruler.
 - a. Work with your partner and use the steps below to construct a line that is tangent to circle S and goes through point A .



Consider pausing the activity after question 2 to bring the class together to ensure that everyone is on the same page with the correct understanding of the concept. A misconception here could cause errors later in the lesson.

10.H1.2 Collaboration: Tangent Lines of Circles (continued)

1.	Use the ruler to draw a line from the center, point S , to the given point .
2.	Construct the perpendicular bisector of $\overline{SA}$ to find its midpoint.
3.	Set the compass width to the distance from the midpoint of $\overline{SA}$ to the circle's center, point S .
4.	Keeping the compass at the midpoint, mark an arc across the circle at two places, creating M and V .
5.	Draw the two tangent lines from A to M and A to V .



To complete question 3, students will need to have a compass and a ruler available. Consider having these prepared and distributed prior to this question in the activity to maintain instructional momentum.

4. Jamie wrote the steps for constructing a line tangent to a circle on note cards, including the steps of the perpendicular bisector, but the cards got out of order.

a. Work with your partner to order the steps 1 – 8.

6	Place the compass on the midpoint of the line segment and adjust its lengths to reach the center.
2	Place the compass at one end of the line segment and adjust the compass to slightly longer than half the line segment.
5	Place the ruler where the arcs above and below the line cross, and draw the perpendicular line.
7	Keeping the same width and compass point at the midpoint, draw a circle.
3	Draw arcs above and below the line.
8	Where the new circle crosses the original circle, create a point and draw the tangent line.
1	Draw a line connecting the center of the circle and the point outside the circle.
4	Keeping the same compass width, draw arcs above and below the line from the other end of the line segment.



Consider showing students the digital version of the step-by-step construction process for a visual element.

10.H1.2 Collaboration: Tangent Lines of Circles (continued)

- b. Explain how Jamie's method is different.

Answers will vary. Jamie drew an entire circle for step 7 and uses that circle for step 8, while the steps in problem 3 only draws 2 arcs that cross the circle.

- c. Explain why Jamie's method works.

Answers will vary. Jamie's method meets the same conditions as the method in problem 3.



Consider adding a layer of engagement by taking these steps in 1-8 and printing them onto colored paper and cutting them into strips. This will provide a kinesthetic element and allow students to sort the steps around and place them into the correct order.



At the conclusion of this activity, engage students in a discussion about their answers to question 4 parts b and c. It is important for students to understand the various methods and why they work.

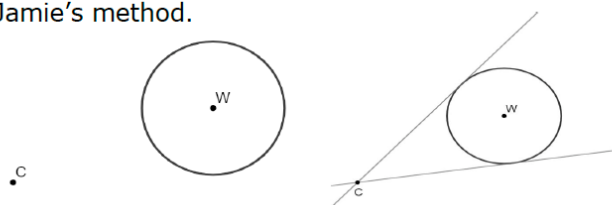
10.H1.3 Your Turn!

Component Overview: Students will independently complete two tangent construction practice problems.



10.H1.3 Your Turn!

- Given circle W and point C outside the circle, construct a line tangent to circle W going through point C using Jamie's method.



- Naomi and Xavier were practicing constructing a line tangent to circle B going through point A , but their results did not match.

Naomi's Construction	Xavier's Construction

- Who correctly drew a line tangent to circle B going through point A ?
Xavier
- Explain the mistake the other student made.
Answers will vary. Naomi's construction does not go through point A.



Consider using this time to work with individual students or small groups based on those who were struggling during the 10.H.2 Collaboration.



Have students who finish early swap their constructions from question 1. Students will then evaluate the classmate's work and offer suggestions for improvement or correction.

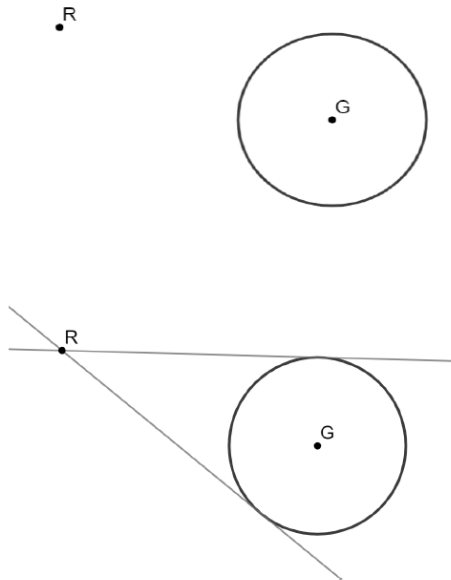
10.H1.4 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door with constructing a line tangent to a circle.



10.H1.4 Wrap-Up: Ticket Out the Door

- Given circle G and point R outside the circle, construct a line tangent to circle G going through point R .



Students construction on this activity can provide valuable data as to whether students understand the content of this lesson.